PARK JUNGHO (Kyle) - CV

Backend Engineer for Quantitative Trading and Blockchain Systems

Backend engineer building cross-exchange arbitrage, on-chain trading, and cross-chain bridges in Rust, Go, and TypeScript. At Kronon Labs, owned 8-exchange order connectivity, PnL, and delta management in the team's arbitrage system. In 2026, joined Mantle, a global company, as Developer Relations Engineer and built the Korea team's content pipeline.

CONTACT

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- **GitHub**: https://github.com/kyle-park-io · **Portfolio**: https://jungho.dev
- **Quant Career**: https://jungho.dev/quant · **Personal Trading Strategies**: https://jungho.dev/personal-quant
- **Live Quant Trading Dashboard**: https://kyle-quant.xyz

PROFESSIONAL EXPERIENCE

Mantle

Developer Relations Engineer | Apr 2026 - Oct 2026 (7 months)
Global institutional infrastructure for real-world assets on the Ethereum L2 Mantle Network
Tech Stack: TypeScript, Next.js, PostgreSQL, Solidity

- **Owned Korea builder onboarding for The Turing Test Hackathon 2026**: a $120K two-phase global flagship hackathon. In Phase 1, the ClawHack AI trading agent contest, ran an X live workshop on building agents and recruited Korean whitelist participants over Telegram. A Korean participant's agent won the volume track at $10.31M, and 12 tracked Korean wallets made up 26.88% of volume. Reported Korean results from the leaderboard API to the global team with Phase 2 requests. Over the 6 weeks of Phase 2, ran an offline co-working session (48 registrations), a live online workshop, and the Seoul hacker house workshop and fireside chat (100+ builders). Presented the hackathon at The Age of Agentic AI meetup, the Mantle Seoul workshop, and the AI community Solp Club
- **Ran the Q402 builder workshop on agent payments (Mantle × Quack AI, Aug 2026)**: 42 builders connected Q402 to their own AI agents over MCP and paid on Mantle mainnet from the chat window, with no wallet app, no signing prompt, and no MNT for gas. Built the operations site as the first version of pay-lab: wallet registration, live leaderboard, quest verification, and a QR payment bridge, with badges and quests judged from on-chain data. 488 on-chain payments in three hours
- **Designed, built, and deployed the Korea team's 16-module content operations pipeline**: X and Lark collection, Korean translation, per-channel conversion including press releases, two-stage human review, and Telegram and X delivery. A reviewer who never opens a terminal approves the copy and sends it from the same screen. From July 2026 the Korean official account's posts were translated and reviewed in it before posting; by October it had collected 446 X posts and made 162 translations and 837 channel versions
- **Ran the Korean official account**: 279 posts, 246K impressions, and a 1.74% engagement rate over Apr-Sep 2026, 1.7× the global account's 1.03%. Korea-team campaigns and event notices took the top six posts by reach

- **Taught a stablecoin payments class at Inha University (Oct 2026)**: a session of the Digital Assets and Market Infrastructure course for finance students, on how stablecoin payments move and what changes when an AI agent pays. Ran two labs on pay-lab: students made a wallet on their phones, sent testnet USDC to a classmate and read the receipt on a block explorer, then signed a spending limit for an AI agent in claude.ai or Codex and watched a request past the limit get refused by both the lab server and the USDC contract. Covered Korea's pending won-stablecoin bills issue by issue: who may issue, reserves and redemption, and interest. Link to the slides, in Korean: https://pay-lab-inha.vercel.app/slides
- **Wrote the Korean technical documentation set (5 documents)**: Mantle block explorer (beginner and deep-dive editions), a DEX comparative deep dive, an AI agent trading path guide, and an asset-withdrawal guide. The deep-dive edition covers modular architecture, Rollup vs. Validium classification, L2's dual gas structure, and why Mantle sits at L2Beat Stage 0. Link to the documents: https://kyle-park.notion.site/Docs-3675e2b105168061b410fdf801eeaac8
- **Co-hosted the Korean X livestream and presented its technical session (Apr - Sep 2026)**: co-hosted from April, ran the Phase 1 and 2 hackathon segments with ClawHack agent traders as guests, and from July presented a technical session monthly. July's RFQ session set AMM, on-chain CLOB, RFQ, and intent execution side by side and became a standalone X article. August's bridge session covered Lock & Mint against Burn & Mint and Mantle's bridge routes: the canonical bridge, LayerZero OFT, and Chainlink CCIP. September's RPC workshop compared seven Mantle RPC providers and priced three of them per month by app type. The August live post drew 18.0K views, the Korean account's second-highest over Apr-Sep 2026
- **Campus tour of university blockchain clubs (May 2026)**: presented Mantle's builder programs and hackathon tracks at seven clubs over three weeks: KAIST, Korea University, Sungkyunkwan, Ewha Womans, Ajou, Inha, and Kwangwoon. Ran a Mantle testnet contract deployment lab at Sungkyunkwan and a live build of a stablecoin pair-trading bot at Ewha. The Korea University, Ewha, Sungkyunkwan, and KAIST clubs posted session recaps on their X accounts. In the research contest introduced on the tour, a Sungkyunkwan club member won the grand prize and the Kwangwoon club the most-participating-club award

Kronon Labs Co., Ltd.

Trading System Backend Engineer | Apr 2025 - Nov 2025 (8 months)
Tech Stack: Go (Fiber, Fx, gRPC), Python (Django), PostgreSQL, MongoDB, AWS (ECS, ECR, EC2, Parameter Store, KMS), Docker

- **CEX Cross-Exchange Arbitrage System (Rothchild)**: a Taker-Taker and Maker-Taker arbitrage system built with the team. Its Go strategy engine trades spreads across Binance, Bybit, Bitget, KuCoin, MEXC, Gate.io, and Bithumb spot markets, Bybit against Jupiter, Bithumb and Upbit KRW markets against Bybit, and the perpetual DEXs Paradex, Backpack, Extended, and Rails. Owned the console backend for orders, balances, and PnL: added order replacement and lookup methods to the common exchange interface a teammate started, and built order placement, cancellation, replacement, and lookup, user-data WebSockets for fills, and keepalive and reconnection for 8 exchanges including Gate.io, plus Solana DEX. Built PnL with 1- and 5-minute history and 1h, 4h, 1d, and 7d totals, a delta manager that nets each asset's delta and clears it on the exchange holding the largest balance, and a dashboard tracking PnL and positions in real time. Set up recorded exchange responses to test each integration
- **Solana DEX On-Chain Trading Bots**: bots in Go running three alpha strategies (New Pair Detection, Back From The Dead, Fat LP Analysis). GoPlus security checks and top-30-holder and fresh-wallet-share entry filters, Jupiter swaps and trigger orders, a price cache with 5-minute and 1-hour volatility, tiered sell rules, and Helius WebSocket monitoring of Pump.fun, Raydium, Orca, and Meteora pools. A Solscan call coordinator built on a lock-free ring buffer and a worker pool queues calls at three priorities and enforces the rate limit in one place. 186 test functions and 25 benchmarks

- **News and KOL Signal Trading Engine**: a futures trading engine built with the team. It ingests private Telegram channels and X list tweets, scores each item's similarity to known events with the average of several OpenAI assistants, and routes signals to Binance and Bybit USDT-futures orders. Built the X list-tweet collector, the private Telegram channel subscription, and the multi-assistant scoring. Evaluated the LLM scoring on a 100-item test list built for it and iterated three versions of the assistant instructions
- **Exchange Listing Sniper**: built with the team, a sniper that catches Upbit and Bithumb listing announcements and sends new symbols over gRPC to order servers in Tokyo and Singapore that place Binance futures, MEXC, and Bybit orders. Ran the earlier listing detector, which watched Upbit and Bithumb notices and ordered on Binance futures, Bybit, MEXC, and OKX, in a dev environment and listed seven improvements
- **Trading Metrics Collection & Analytics**: for an altcoin performance chart backend built with the team, which ranks the top 50 altcoins against BTC alongside open-interest-to-market-cap ratios from Velo, built the Bybit 1-minute kline WebSocket collector and kline API and added time ranges and BTC change rates to the performance API. Also designed a batched metrics collector for strategy detectors
- **Prediction Market Monitoring (Totoro)**: with the team, a module that pairs Polymarket and Kalshi NFL and Premier League markets by match and pulls their prices

Medium Inc.

Backend Engineer | Sep 2022 - Feb 2024 (1 year 6 months)
Tech Stack: Go, TypeScript (NestJS), Solidity, Hyperledger Fabric 2.2, Kafka, MySQL, AWS, Docker

- **Cross-Chain Bridge Infrastructure**: built with the team, a multisig-verified bridge linking the company L1 mainnet RIGO (Cosmos Tendermint with EVM), the private consortium chain MDL (a Hyperledger Fabric 2.2.2 fork), and Ethereum. A Kafka event pipeline, 17 bridge chaincodes, and Solidity contracts
- **Security Token Platform (ERC-1400)**: an STO token designed and built as a Go chaincode on Hyperledger Fabric to the ERC-1400 interface: partition-based issue, transfer, and redeem, and token wallets
- **RIGO Chain Communicator**: a Go client sending RIGO transactions over Tendermint RPC, deployed in-house
- **Hyperledger Fabric Network Automation**: Bash scripts that automate Fabric CA enrollment, genesis block and channel transaction generation, orderer, peer, and CouchDB startup, channel creation and joins, anchor peer updates, the chaincode package, install, approve, and commit steps, and the block explorer connection. Seven Docker Compose configurations with different channel, organization, and orderer counts on Fabric 2.2.6 and 2.2.13 images, the largest with 3 organizations and 2 etcdraft orderers, plus four configurations that run the same sequence from Fabric binaries on Linux and macOS without Docker

BF Labs Inc.

Backend Engineer | Aug 2022 - Sep 2022 (2 months)

- **Government Research Project**: built the report intake and lookup REST API server for a project funded by Korea's IITP with Node.js, Express, and MariaDB

EDUCATION & LANGUAGES

- **Kyung Hee University, Seoul**: Bachelor of Engineering in Mechanical Engineering | Mar 2014 - Aug 2022
- **English**: TOEIC Speaking 150/200, Intermediate High · **Korean**: Native

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TECHNICAL SKILLS

- **Specializations**: CEX/DEX Trading Systems, Exchange Connectivity and Order Management (REST, WebSocket, FIX/SBE), PnL/Delta Management, Cross-Exchange & Triangular Arbitrage, DeFi Protocols, Cross-Chain Bridges, Smart Contracts (ERC-4626, ERC-1400), Workflow Automation, Technical Writing
- **Languages**: Rust, Go, Python, TypeScript
- **Blockchain**: Solidity (Foundry, Hardhat), Solana (Rust), Hyperledger Fabric
- **Backend Frameworks**: NestJS, Next.js, Fastify, Django, Gin, Fiber, Fx
- **Infrastructure**: AWS, GCP, Docker, Kubernetes, PostgreSQL, MySQL, Kafka, Redis, Vercel
- **AI Tooling**: Claude Code, Codex, MCP servers, agent skills, trigger evals, guardrail hooks, human-in-the-loop pipelines
- **Frontend**: React, SolidJS, Astro

SELECTED PROJECTS

mantle-kr-herald: Human-in-the-Loop Multi-Channel Content Pipeline

Tech Stack: TypeScript, PostgreSQL, Vercel, Vitest, Claude Code

- **Re-runnable stages**: each stage from collection to delivery is its own command, approval is kept apart from sending, and a per-item lineage records every stage's output so each review edit can be traced
- **Translation loop**: a local coding agent fills a worksheet and a human approves it. Approved translations become few-shot examples, a translation memory of paired official English and Korean posts adds precedents with overlapping tags, and weekly glossary mining from human edits drops one-off corrections the Korean account never uses. The 178-term glossary feeds the prompt and a term check
- **Repurposing and multi-channel delivery**: one approved translation becomes 12 content types such as announcements, explainers, KOL copy, and press releases, formatted for 9 channels and sent via the Telegram Bot API and Typefully to X, LinkedIn, Threads, Bluesky, Mastodon, and Substack, with a unique index on the delivery ledger for idempotent sends
- **Config split and hexagonal structure**: accounts, glossary, and conversion guides live in config, so another brand can reuse it. With external APIs and storage behind 34 ports, moving storage from files to PostgreSQL on Vercel serverless added 19 adapters without touching domain code. 5,115 test cases; storage adapters run against in-process Postgres

pay-lab: On-Chain Payments Lab Kit Reconfigured per Event

Tech Stack: Next.js, PostgreSQL, Privy, viem, MCP, Foundry, Vercel

- **Reusable kit**: rebuilt the Q402 workshop site into lab modules, per-event config, and chain adapters, then ran the Oct 2026 Inha University class on it; an architecture test enforces layer dependencies
- **Gasless payments and agent limits**: relays students' EIP-3009 signed USDC transfers and pays the gas. Agent wallet tools spend only within the student's EIP-2612 permit, through a remote MCP server on wallet-signature OAuth for claude.ai and Codex that takes the sender only from the access token
- **Money-path safety**: funding reserves a record and sends at most once, and stalled records are reconciled against the chain. A 30-student load test found Vercel functions reusing a nonce and failing 3 to 8 of 30 transactions, fixed with a PostgreSQL advisory lock and a nonce table that serialize each wallet's sends

Binance Triangular Arbitrage Engine

Tech Stack: Rust, FIX 4.4, SBE (Simple Binary Encoding), rustls

- Rust engine that logs on to the Binance FIX API with an Ed25519 signature and opens market data and order entry sessions in SBE binary encoding, keeping order books from snapshots and incremental refreshes and handling new orders, cancels, and execution reports
- On every book update, prices both directions of the BTC, ETH, and USDT triangle after each symbol's taker fee and keeps the direction that clears the threshold. Runs as a dry run by default, with Telegram alerts

Polymarket 15-Minute Market Complete-Set Arbitrage Bot

Tech Stack: Go, Rust

- Buys both outcomes in Polymarket's 15-minute crypto Up/Down markets when their best asks sum below $1. Once both legs fill, each pair is worth $1 at settlement or on a merge; fees and leg risk decide whether it pays
- A stale-quote filter drops a signal when the two quotes are over 29ms apart, the older one is over 29ms old, or over 29ms passes before the order. A later version re-derived the threshold from the fee table and pre-signs orders across price levels to send both legs as FOK, so no unfilled order rests on the book. Execution is off by default, with Telegram alerts

trustline: Off-Chain Security Scanner for Web3 Projects

Tech Stack: TypeScript, Node.js, GitHub Actions · https://github.com/kyle-park-io/trustline

- Built after leading the response to a breach, a Next.js server taken through React2Shell (CVE-2025-55182). **9 scan modules** check the off-chain stack before deploy for the mistakes that let one breach spread: secrets in build output, management ports open to the internet, over-privileged cloud IAM, account-wide credentials, persistence artifacts, and disabled audit logs
- Published on **npm** (`npx trustline-scanner scan .`) and as a **GitHub Marketplace Action**, with SARIF output for GitHub code scanning. A read-only on-chain command recovers the EtherRAT malware's command-and-control values from Ethereum smart contracts and maps the contracts around them; the findings feed the scanner's IOC set

skills: Agent Skill Marketplace and Guardrail Hooks for Claude Code and Codex

Tech Stack: Agent skills, Python, Bash

- Turns repeated engineering decisions into skills in **7 domain plugins**, with Claude Code and Codex manifests generated from one source and stack presets that start a new repository with the right skills on
- **Trigger evals**: runs Claude Code and Codex non-interactively with the plugin installed and watches the session stream for the target skill. Negative queries belong to a neighbouring skill; project-templates scored 1.0 precision and recall over 14 queries run 3 times each
- **Guardrail hooks**: PreToolUse hooks block nested subagents, more than 5 spawns in 10 minutes, unconfirmed parallel sessions, .env secret output, and banned writing patterns, with cost caps set after a delegation grew to 21 subagents in 3 minutes

writing-guides: Instructions and Output Checks for Agent-Written Copy

Tech Stack: Markdown, Node.js, Python

- 53 instruction files for agent-written Korean copy, screen copy, translation, talk scripts, and documents. Each rule cites the feedback, post, or measurement behind it, and corrections update both
- The document checker and the agent PreToolUse hook share banned patterns, checked for parity before commit; a deck toolchain builds PDF, HTML, script, and slides from one master and checks contrast and layout

muster: Local Scheduler Dashboard

Tech Stack: React, Vite, Tailwind CSS, Fastify · https://github.com/kyle-park-io/muster

- Shows Windows Task Scheduler, systemd timers, cron, and Codex and Claude schedules per environment on one screen and toggles or runs user-owned jobs. Bound to 127.0.0.1; every change shows the exact command first

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AWARDS & CERTIFICATIONS

- **1st Place, DeFi & Composability Track, and Mantle Incubation Grant, Mantle Global Hackathon 2025** (DOOR Protocol) | Feb 2026
  - 500+ submissions from 2,000+ builders across 6 tracks and $150K in prizes. DOOR Protocol won its track and was one of 3 teams picked for Mantle incubation grants that continue after the event
  - A DeFi rate protocol on Mantle that splits yield and risk into senior and junior tranches. Entered as a team and wrote the 7 core contracts (150 test functions) for ERC-4626 tranche vaults, a safety module, withdrawal epochs, and a rate oracle, plus the Next.js dashboard and keeper bot
- **3rd Place, Seoulana Hackathon 2025** (Solana blockchain development competition) | Apr 2025
  - Built the Orca Whirlpool pool, position, and liquidity transactions for the entry, Stream Square
- **Engineer General Machinery** (Human Resources Development Service of Korea) | Dec 2020

LEADERSHIP & RESEARCH

Orakle: KAIST-based Blockchain Research Society

Vice President (8th Cohort) | Mar 2026 - Present · **Team Leader, ODA Team (7th Cohort)** | Sep 2025 - Feb 2026

- Society operations and program direction. Organized the 8th cohort's final research conference (Sep 2026, Hana Bank Lounge, Samseong)
- **Wrote the 7th cohort's five published DeFi research papers** (CC BY 4.0): https://github.com/orakle-7th-sda/conference-pdf
  - Stablecoin depeg survey covering CEX cases from 2018 to Feb 2026 and on-chain cases from 2022 to Feb 2026, by venue, magnitude, cause, and duration
  - Solana searcher/solver strategy design. Six strategy families for real-time depeg arbitrage on Jito bundles, Jupiter v6, and Helius WebSocket
  - Plus a 17-category DeFi ecosystem taxonomy and a cross-category stablecoin map
- **6th Cohort BTCFi Bitcoin Options Service** (May 2025 - Nov 2025): one-way option PoC on BitVMX-CPU, a Go backend (60 REST APIs, WebSocket, PostgreSQL), and the BitStriker frontend (React, MutinyNet wallet)